Automatic Stand Mixer

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Abstract**—** The Automatic Stand Mixer is an innovative kitchen appliance designed to simplify and enhance the cooking experience through automation and real-time monitoring. Built around an Arduino microcontroller, the system integrates a keypad, IR remote, LCD display, servo motor, DC motor, RGB LED, and a buzzer to provide a seamless and user-friendly mixing solution. The mixer offers adjustable speed settings (50%, 75%, and 100%) and timed mixing modes (ranging from 30 seconds to 6 minutes), ensuring versatility for various recipes. Visual and auditory feedback mechanisms, including an LCD display, RGB LEDs, and a buzzer, keep users informed about the mixer's status, while a servo motor automates the beater's movement for added convenience. The prototype demonstrates the effective integration of hardware and software, showcasing the potential of embedded systems in domestic applications. Beyond its functional benefits, the Automatic Stand Mixer serves as an educational tool, illustrating the practical application of microcontroller-based automation. This paper discusses the design, implementation, and performance of the prototype, highlighting its role as a time-saving device and a teaching tool for advancing knowledge in automation and embedded systems.

Index Terms**—** *Automation. Embedded Systems, Microcontroller, Arduino, Kitchen Appliances, Real-Time Monitoring, Servo Motor, DC Motor, LCD Display, IR Remote Control, Keypad, PWM, Educational Tool, Home Automation*

I. Introduction

Interactive automation tools are more commonly implemented in domestic and industrial environments to optimize convenience, efficiency, and safety. Based on embedded systems, microcontrollers, and real-time monitoring, these tools present new solutions while further advancing users' knowledge of technology through touch operation [1]. Kitchen automation is a characteristic area of implementation where new technology simplifies conventional tasks for intelligent and more efficient cooking processes [2].

The Automatic Stand Mixer is a prime example of this, combining real-time monitoring, control, and automation into one convenient design. It has an LCD I2C display for status, LEDs and buzzer for notifications, and an IR remote control /Keypad for effortless use. A servo motor takes care of automating beater movement, and a DC motor provides adjustable speed mixing for accuracy and efficiency.

Built for diverse users, the system has adjustable settings for speed, time, and controls for operation, hence being both functional and pedagogical. By integrating embedded systems with practical uses, the project illustrates how automation can make life easier while also being a teaching tool. The paper discusses its design, implementation, and performance, with a focus on its use as a time-saving device and an educational tool for microcontroller-based automation.

II. Methodology

The Automatic Stand Mixer will be developed by applying concepts in Real-Time Embedded System.

A. Hardware Development

Automatic Stand mixer components consist of materials that are readily available and cheap for easy recreation and development. Automatic Stand Mixer uses several components shown in figure 1. ; Keypad/IR Receiver & Remote for user input, LED, LCDI2C, buzzer for user interface and status indicators, Servo Motor for automatic lifting of the beater, and Gear Motor for the mixing mechanism.

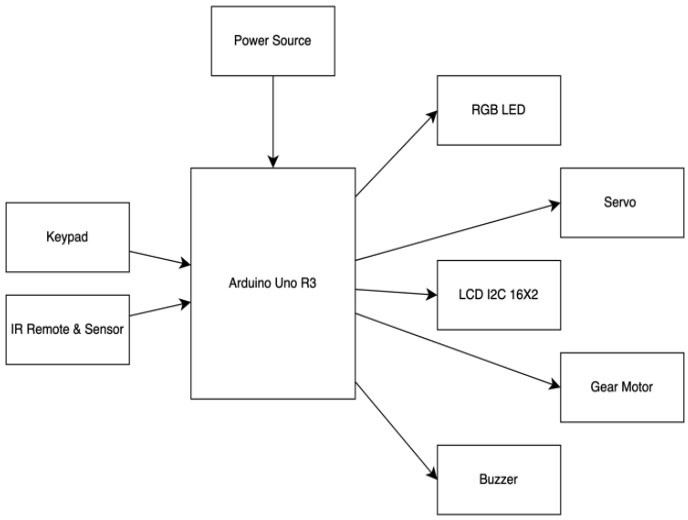

Figure 1: Hardware Block Diagram of Automatic Stand Mixer

The IR receiver or keypad and remote allow the user to input commands like setting the mixing time, halting the process, and changing the mixing speed. The LCD I2C module gives real-time feedback on the status of the mixer, showing crucial information such as the present speed level. The RGB LED acts as an indicator of status, where colors change to denote varying speed states—i.e., 50%, 75%, and 100%—enabling users to track the operation of the mixer with ease.

The servo motor controls automatic beater rising or falling, reducing manual intervention. The gear motor drives the mixing drive, which enables smooth and continuous motion for effective mixing performance. Moreover, the buzzer also gives audible feedback to inform users of state changes, faults, or the successful completion of the mixing cycle. All of these elements are controlled by the Arduino Uno R3, which handles user inputs and manages the different actuators to automate the mixing process effectively.

B. Software Development

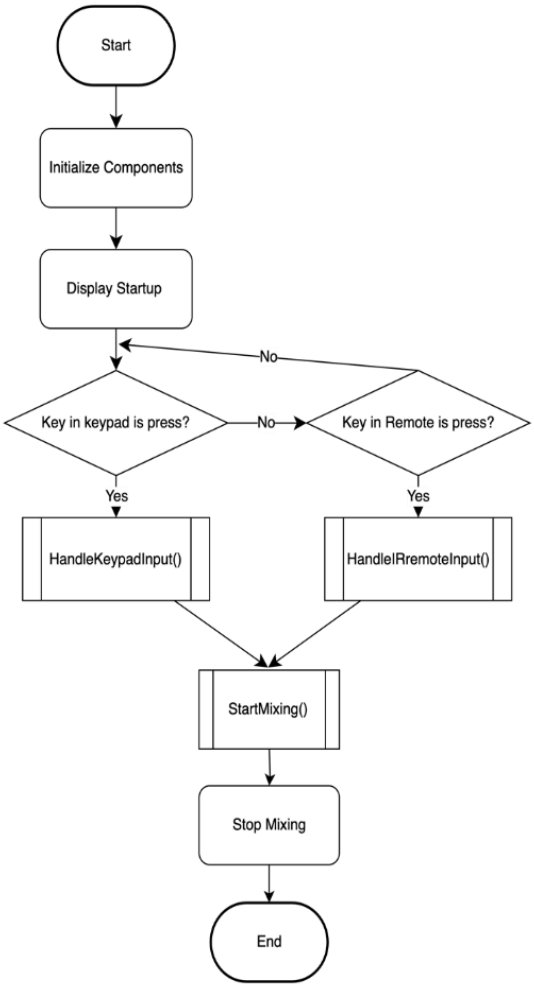

Figure 2.1: Main Flowchart of the Software Development of the Parity Panic

Figure 2.1 illustrate the main flowchart of the whole system wherein the system will initialise and display startup configuration. This includes opening the backlight of the LCD and starting I/O capabilities of each component. After which, handling the inputs from either keypad and IR remote is done allowing the startmixing function to start the motor with right speed, time, and right indicator. Once certain time and condition required is achieved the system will loop back to checking of keypad/IR remote input continuously until the program breaks.

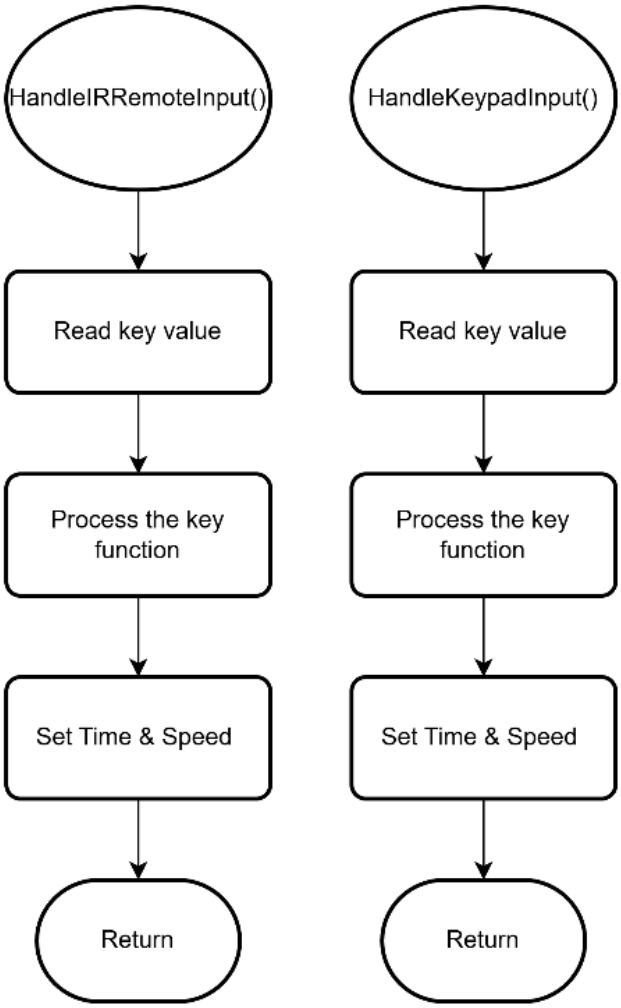

Figure 2.2: Inputs Handle Flowchart

Figure 2.2 shows the predefined function HandleKeypadInput() & HandleIRRemoteInput() whole logic. The main process involves setting different speeds and durations for the mixer, as well as controlling an LED indicator and a servo motor for tilting.

When a key is pressed or an IR signal is received, the program checks the value of the input and executes corresponding actions. For example, pressing keys '0' through '6' on the keypad starts the mixer with predefined mixing times ranging from 30 seconds to 6 minutes. During the mixing process, the speed of the mixer is adjusted in stages (100%, 75%, or 50%) based on the elapsed time. The speed is controlled using PWM (Pulse Width Modulation) via the analogWrite function, and the mixer's motor is activated by setting the IN1 pin to HIGH. Additionally, an LED changes color to indicate the current speed state (red for 100%, yellow for 75%, and green for 50%). A servo motor (tiltServo) is also adjusted to a fixed position (40 degrees) during mixing.

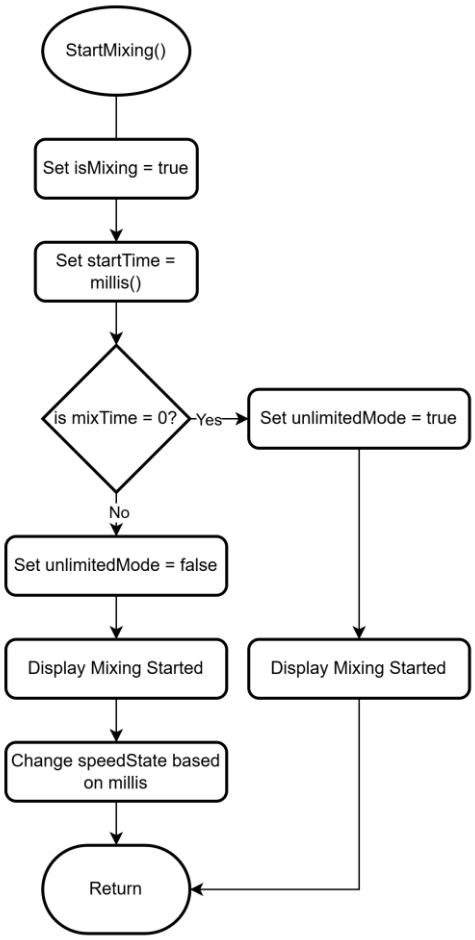

Figure 2.3: StartMixing Function Flowchart

Figure 2.3 represents the startMixing() function, which initiates mixing by setting isMixing to true and recording the startTime. It checks if mixTime is zero to enable unlimitedMode; otherwise, it runs for a set duration. The LCD displays "MIXING STARTED" with the speed percentage, and the LED color changes based on speedState (Green for 50%, Yellow for 75%, Red for 100%). Finally, the function updates speedState using millis() and returns, ensuring real-time mixing control.

Theses flowchart illustrates the key processes in the Automatic Stand Mixer system. Figure 2.1 outlines the main flow, starting with system initialization, enabling LCD backlight, and setting up I/O components. It then continuously checks inputs from the keypad and IR remote to adjust mixing parameters. Figure 2.2 details input handling, where keypresses or IR signals determine mixing speed and duration. The mixer operates at 100%, 75%, or 50% speed, indicated by corresponding LED colors, while a servo motor tilts as needed. Figure 2.3 focuses on the startMixing() function, which sets the mixer in motion, manages unlimited mode, and adjusts the speed dynamically based on elapsed time.

C. Casing

Figure 3: Automatic Stand Mixer Casing

The casing for the Automatic Stand Mixer is crafted from cardboard, designed to mimic the aesthetics of a KitchenAid stand mixer while maintaining a lightweight and functional structure. The LCD display is securely embedded on the front panel for clear visibility of system status, and a keypad is integrated for user input, complemented by an IR remote for additional control. The mixing mechanism, including the whisk, is firmly attached to the upper section, ensuring stable operation. The base provides a designated space for the mixing container, offering balance and support during use. Decorative stickers enhance the overall appearance, adding a unique and playful touch to the design.

III. Results and Discussion

The prototype was successfully developed following the flowchart outlined in Chapter II. This structured approach ensured a smooth integration of functions, allowing seamless updates and effective communication between various variables and components. Figure 4 presents a snippet of the startMixing() function, which plays a crucial role in handling mixing speed and time depending on user input. The design, inspired by KitchenAid stand mixers, features a custom-built casing made from cardboard, giving it a nostalgic yet functional appeal. The LCD screen displays real-time updates on the selected speed and mixing status, while the keypad and IR remote provide users with flexible control options. Additionally, an LED indicator system visually represents different speed levels, and the motorized whisk mechanism ensures efficient mixing, making the device both intuitive and user-friendly.

```
void startTimedMixing(unsigned long duration) {
  startTime = millis();
  mixTime = duration;
  isMixing = true;
  unlimitedMode = false;

  while (millis() - startTime < mixTime) {
    unsigned long elapsedTime = millis() - startTime;

    if (elapsedTime < mixTime / 3) {
      setSpeed(100);
    } else if (elapsedTime < 2 * mixTime / 3) {
      setSpeed(75);
    } else {
      setSpeed(50);
    }

    tiltServo.write(40);
    updateLCD();
  }
  stopMixing();
}

void setSpeed(int speed) {
  speedState = speed;
  int pwmValue = PWM_VALUES[(speedState / 25) - 2]; // Map speed to PWM value
  analogWrite(ENA, pwmValue);
  digitalWrite(IN1, HIGH);
  setColorBasedOnSpeed();
  updateLCD();
}
```

Figure 4.1: startMixing() Function

The servo motor reliably tilts the mixer head to 40° during operation and returns it to 15° when idle, demonstrating precise control, and the DC motor operates smoothly across all speed settings, with no noticeable lag or inconsistency. The prototype was tested under various conditions, including different mixing durations and speeds (figure 4.2 – 4.4), and all functions performed as expected, with no hardware or software failures. User feedback highlighted the system's ease of use and reliability, particularly the intuitive controls and clear visual/auditory feedback. The automatic stand mixer prototype demonstrates the effective integration of hardware and software to create a versatile and user-friendly kitchen appliance.

Figure 4.2: Unlimited Time Mode (Speed 100%)

Figure 4.3: Unlimited Time Mode (Speed 75%)

Figure 4.4: Unlimited Time Mode (Speed 100%)

The timed mixing modes, which automatically adjust speed at predefined intervals, mimic the functionality of high-end commercial mixers. This feature not only enhances the mixer's usability but also ensures consistent results for various recipes.

The visual and auditory feedback mechanisms (LEDs, LCD, and buzzer) further improve the user experience by providing clear and immediate status updates. Figure 4.5 – 4.7 displays prototype visual when timed mixing mode is selected.

Figure 4.5: Time Mode (Speed 100%)

Figure 4.6: Time Mode (Speed 75%)

Figure 4.7: Time Mode (Speed 50%)

Overall, the automatic stand mixer prototype successfully demonstrates the feasibility of using an Arduino-based system to create a versatile and user-friendly kitchen appliance, providing a robust foundation for further development. With additional refinements and features, this prototype has the potential to evolve into a commercially viable product.

IV. Conclusions and Recommendations

The Automatic Stand Mixer prototype successfully demonstrates the integration of embedded systems, real-time monitoring, and automation to create a functional and user-friendly kitchen appliance. By combining hardware components such as an Arduino microcontroller, keypad, IR remote, LCD display, servo motor, DC motor, LEDs, and a buzzer, the system provides a seamless and efficient mixing experience. The prototype offers adjustable speed settings (50%, 75%, and 100%) and timed mixing modes (ranging from 30 seconds to 6 minutes), ensuring versatility for various cooking tasks. The inclusion of visual and auditory feedback mechanisms, such as the LCD display, RGB LEDs, and buzzer, enhances user interaction and ensures clear communication of the mixer's status. The servo motor's precise control over the beater's movement and the DC motor's smooth operation across all speed settings further highlight the system's reliability and performance. The project also serves as an educational tool, illustrating the practical application of microcontroller-based automation in everyday life. By simplifying conventional kitchen tasks and introducing users to interactive automation, the Automatic Stand Mixer bridges the gap between technology and domestic utility. The successful implementation of the prototype, validated through rigorous testing and positive user

feedback, underscores its potential as both a time-saving device and a teaching tool for embedded systems.

While the prototype performs well, there are several areas for improvement and future development. First, the user interface can be enhanced by implementing a menu-driven system on the LCD display or adding a touchscreen for more intuitive control. Second, safety features like thermal sensors and overload protection should be added to prevent overheating and damage to the motor. Third, pre-programmed mixing profiles for common recipes could simplify the cooking process further. With continued development and refinement, the Automatic Stand Mixer has the potential to become a staple in modern kitchens, revolutionizing the way we approach cooking and food preparation while serving as an educational tool to inspire future innovations in automation.

V. References

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